Design report

wb-cdc-analytics · end-to-end analytics engineering pipeline

Senior Data Engineer position assessment | Matiwos Desalegn | August 2026 Repository: `wb-cdc-analytics` | One command: `make demo`

Executive summary

A public REST API is ingested into PostgreSQL, change events are streamed out of the write-ahead log by Debezium into ClickHouse, and dbt shapes a staging layer, an analytics mart and an ML feature table. It starts with `make demo` and verifies itself with `make verify`.

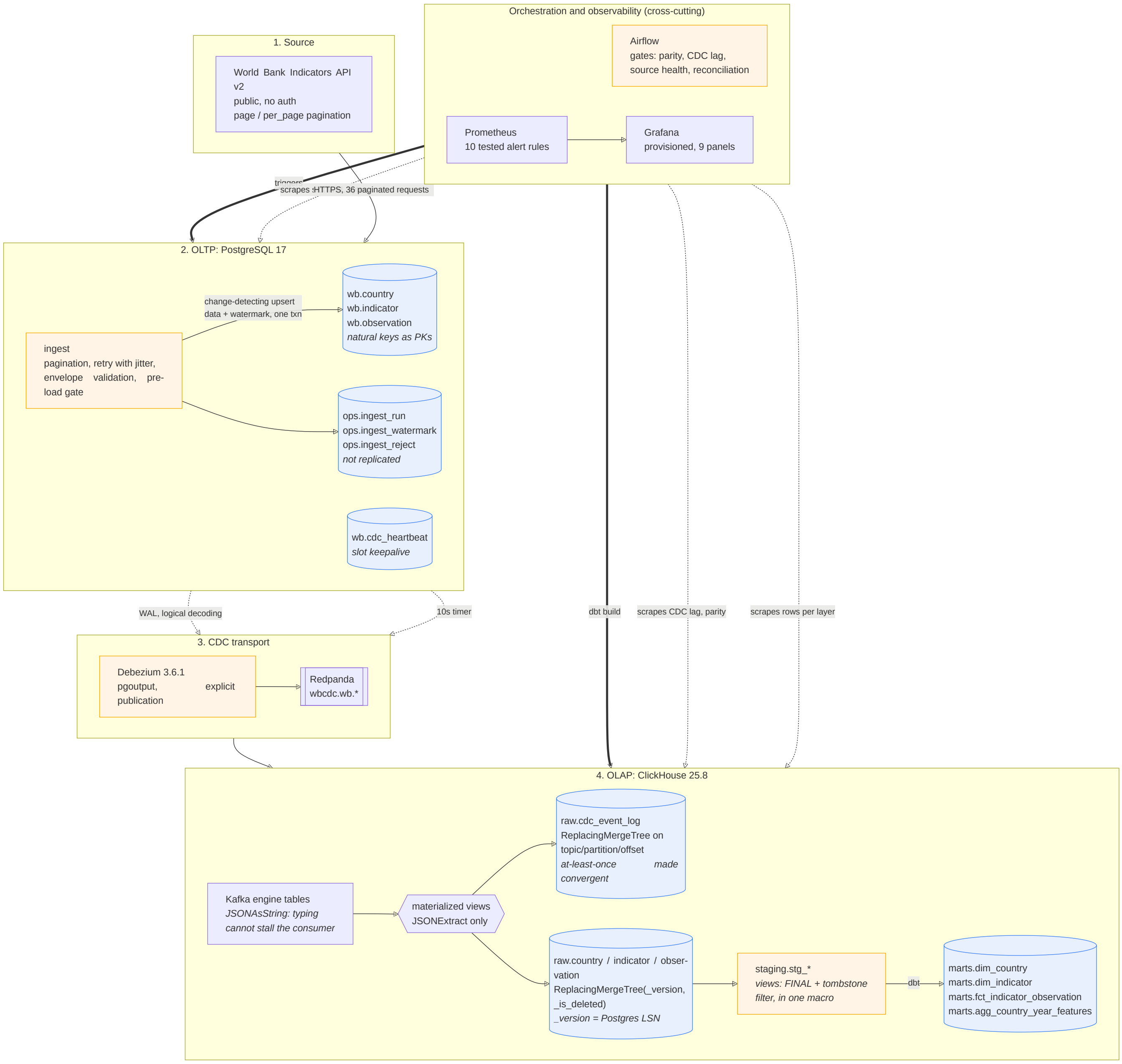
Measured on this stack: 2,970 observations (5 countries, 9 indicators, 66 years); **about 3 seconds** from PostgreSQL commit to a queryable ClickHouse row; 58 dbt tests and 59 unit tests green; 10 Prometheus alert rules, all 10 unit-tested with `promtool` across 13 cases.

Three decisions I would defend, each preventing a failure that produces **no error at all**:

1. **The natural key is the primary key.** Under `REPLICA IDENTITY DEFAULT` a delete event carries only the primary key, so a surrogate would make deletes arrive downstream as an integer with no way to identify what was removed.
2. **The upsert is change-detecting.** Without `WHERE source_hash IS DISTINCT FROM ...`, a re-ingest rewrites every row and each no-op update emits a CDC event, so the change stream describes the scheduler instead of the data.
3. **The Kafka engine reads raw JSON and types in materialized views.** Typed Kafka-engine columns look cleaner and are a trap: a parse failure fails the block, offsets are never committed, and the consumer retries forever in silence.

I optimised for decisions I can defend and for failures that announce themselves. Most of what I have had to debug in production was not a component that crashed but a component that kept running while quietly doing the wrong thing, so wherever there was a choice between a design that reads elegantly and one that fails loudly, I took the second.

Architecture diagram



Architecture. Full resolution: diagrams/exports/. Source: diagrams/src/architecture.mmd

Source: `diagrams/src/architecture.mmd`, also inline in the README. Nine containers at full extent; the **core data path is four** and fits in ~2.5 GB, with observability and orchestration behind Compose profiles. Every image tag pinned, every stateful service healthchecked, dependents gated on `service_healthy` rather than start order.

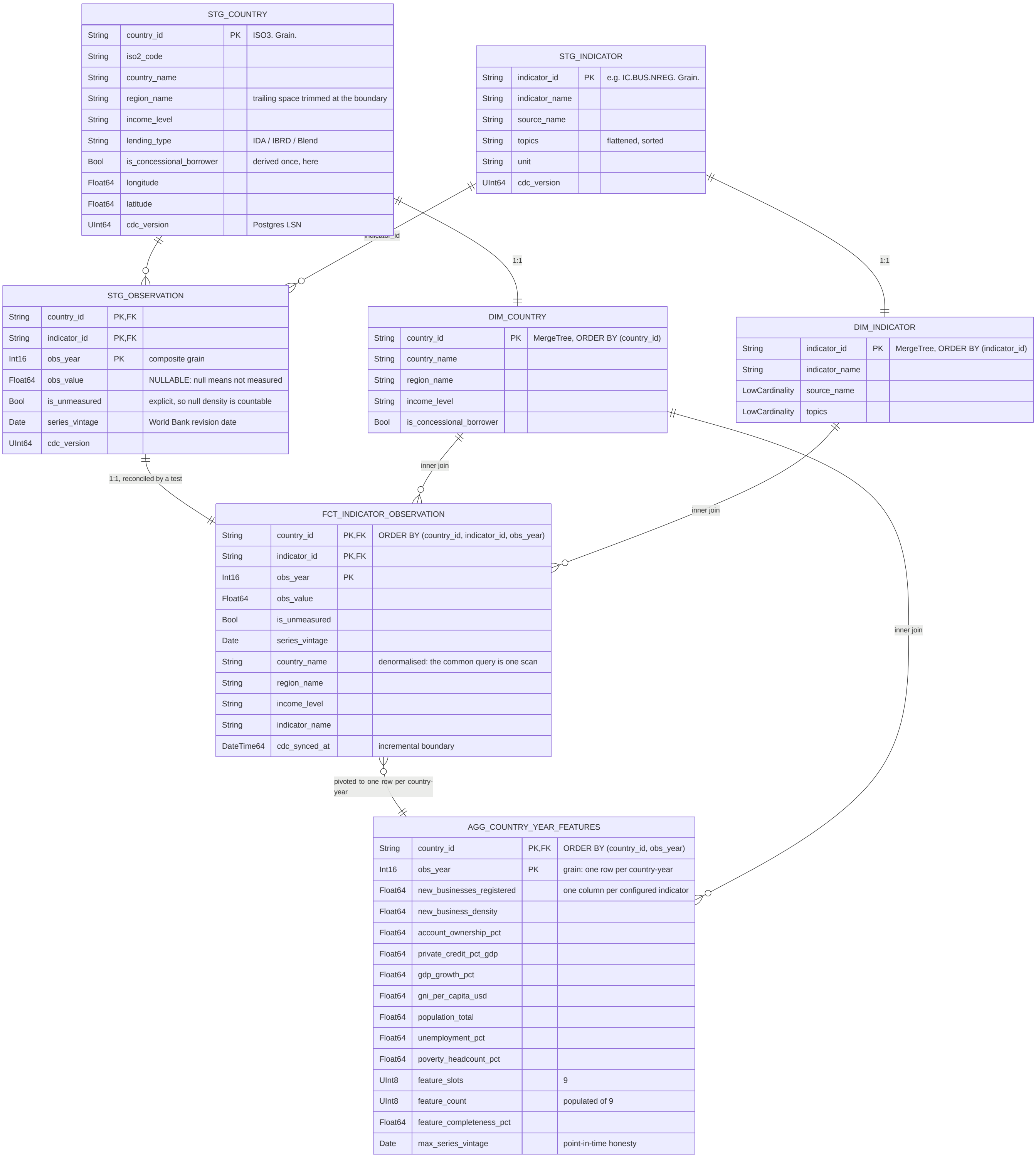
Data flow explanation

Following **one record** end to end, naming the guarantee at each hop. The record is Ethiopia's *New businesses registered* value for 2023.

Hop	What happens	Guarantee
API to memory	`GET /v2/country/TCD;ETH;KEN;RWA;SSD/indicator/IC.BUS.NREG?page=3&per_page=100`. Nine indicators over four pages each, so 36 requests for the fact stream. The page count is taken from the **first** response only, because a page past the end returns a recalculated and wrong one	At-least-once, safe: nothing is written yet, and retries use backoff with jitter
Validation	Typed contract. `countryiso3code` is used, **not** `country.id`, which is ISO2 here. Series vintage attached from page *metadata*. Content hash over business fields only	Fail-closed: a bad row is rejected with a reason into `ops.ingest_reject`, and a mostly-rejected batch fails the run
Load to PostgreSQL	`ON CONFLICT (country_id, indicator_id, obs_year) DO UPDATE ... WHERE source_hash IS DISTINCT FROM EXCLUDED.source_hash`	Idempotent and quiet. Verified: a second run reports `unchanged=2970`. Data, audit row and watermark commit in **one transaction**, so a recorded position can never claim work that rolled back
WAL to Debezium	Logical decoding via `pgoutput`, explicit publication over four tables	At-least-once, positioned by LSN. Explicit publication means adding an unrelated table cannot silently change what streams
Debezium to Redpanda	`ExtractNewRecordState` flattens the envelope, adds `__op`, `__lsn`, `__deleted`, `__source_ts_ms`	`delete.tombstone.handling.mode=rewrite` is what makes deletes **visible** rather than absent
Redpanda to ClickHouse	Kafka engine reads one `String`; two views fire, one to an immutable log, one to typed current state	At-least-once **made convergent**: offsets commit after the block flushes, so a crash re-delivers, and the log's sort key is the Kafka coordinate triple so a duplicate collapses into the row it duplicates
Landing to staging	`FINAL` plus `_is_deleted = 0`, both from a single shared macro	Exactly-once in effect: versions converge to the highest `_version`, which is the source LSN, so "newest" follows the source's commit order, not arrival order
Staging to marts	`dbt build`, `delete+insert` on the natural key, lookback window	Idempotent. Verified: three consecutive builds leave the fact at exactly 2,970 rows

The honest summary: **every hop is at-least-once, and the design makes duplication harmless rather than pretending it cannot happen.** Two mechanisms do that work throughout: a natural key that is stable and present at every hop, and version columns that make "latest" a property of the source rather than of timing.

Data model and schema documentation



ERD. Full resolution: diagrams/exports/. Source: diagrams/src/erd.mmd

Source: `diagrams/src/erd.mmd`. The landing layer mirrors the OLTP schema plus CDC metadata, so the ERD shows staging and marts.

Four layers, each with exactly **one writer**, so "who put this row here" always has one answer: `raw` (materialized views), `staging` (dbt views), `marts` (dbt tables), `ops` (pipeline metadata). Three entities, not thirty: two dimensions plus one fact is the minimum that makes referential-integrity tests meaningful and an ERD worth drawing. **Scale is demonstrated along the indicator axis**, which is a config list, not a schema change.

Table engine selection

Table	Engine	Why
`raw.<entity>`	`ReplacingMergeTree(_version, _is_deleted)`	CDC produces many versions per key. `_version` is the **Postgres LSN**, monotonic per server, so "newest" follows the source's commit order. Arrival time would reorder events under retry
`raw.cdc_event_log`	`ReplacingMergeTree` on `(topic, partition, offset)`	That triple is globally unique per message, so an at-least-once redelivery collapses into the row it duplicates
`staging.*`	View	It only deduplicates, drops tombstones and renames. Storing that adds a second copy that can go stale
`marts.dim_*`, `agg_*`	`MergeTree`	Staging already collapsed the versions, so these have one row per key by construction. A Replacing engine would add merge work with nothing to merge
`marts.fct_*`	`MergeTree`, incremental `delete+insert`	Idempotent re-application on the natural key, which matters because the orchestrator retries

Three `ReplacingMergeTree` traps, all producing wrong data rather than errors. Deduplication happens at **merge time** and merges are asynchronous, so between merges the table legitimately holds several versions and a plain `SELECT` returns duplicates; correct reads need `FINAL`. Second, `SELECT ... FINAL` *does* hide `_is_deleted` rows on 25.8.29 (measured both ways) but does **not** remove them from disk, so any read omitting `FINAL` still sees them. Third, a Replacing engine with no `ORDER BY` collapses the table to **one row**, because dbt-clickhouse emits `ORDER BY (tuple())` when `order_by` is unset.

The control: `FINAL` and the tombstone filter live in **exactly one macro**, and engine plus sort key are always declared together in a model's own `config()` block. A CI convention gate enforces both.

Ordering key

The sort key defines row identity for deduplication, so it is **exactly the source primary key**, no more and no less. Adding a column would stop an update collapsing onto the row it updates; omitting one would merge distinct entities. `raw.observation` and the fact both order by `(country_id, indicator_id, obs_year)`, which is also the access pattern: filter country, then indicator, then range-scan years. No sort key contains a `Nullable` column, because a nullable business key cannot identify a row.

Partitioning key

`PARTITION BY tuple()` **everywhere except the event log: deliberately not partitioned.** Partitioning prunes scans and drops data cheaply *at scale*; at 2,970 rows it does neither. It would create many small parts, push towards the `too_many_parts` threshold (an approaching **write outage**, since ClickHouse refuses inserts past 300 parts per partition), and slow merges, which for a Replacing engine directly slows deduplication.

The adoption trigger is written down rather than left to taste: **partition by month past roughly 100 million rows, or when retention needs whole-partition drops.** The event log *is* partitioned by month, because it is the one table with a TTL and monthly partitions let expiry drop parts instead of mutating rows. A Grafana panel tracks active parts, so the decision is monitored rather than asserted.

Materialized views

Six. A ClickHouse materialized view is an **insert trigger**, not a cached query. Two per topic: one appends to the immutable log, one maintains typed current state. Separate because they have different keys, engines and retention, and because a fault in the modelling view must not stop the log from recording what arrived.

The key decision is that the Kafka engine table has **one** `String` **column** with `kafka_format = 'JSONAsString'`. The obvious alternative, `JSONEachRow` with typed columns, fails brutally: a message that does not fit throws, the block fails, offsets are **not** committed, and the engine retries the same block forever. Nothing inserts, no client sees an error, and the only symptom is a landing table that stopped growing. With `JSONAsString` the consumer cannot fail, typing moves into views using only `JSONExtract*` (which returns a default rather than throwing), and the pipeline degrades to nulls instead of stalling.

Every cast was verified against a live server, and the wire format read off a real topic. Three encodings would have been wrong if assumed: `__deleted` is the **string** `"true"` (so `JSONExtractBool` returns false for a deleted row, resurrecting every delete); a `DATE` arrives as an int32 day count while a `TIMESTAMPTZ` in the same row arrives as an ISO-8601 string; and delete events fill non-key `NOT NULL` columns with **schema defaults**, not nulls. Details: `docs/cdc-wire-format.md`.

Data quality and validation at each stage

Nine mechanisms across six stages, each either a gate in the DAG's critical path or a test in `dbt build`.

Stage	Mechanism	Catches
API response	Envelope **shape** validation, not status code	An error returned with HTTP 200. Ingesting zero rows looks identical to an indicator with no data
API response	Per-indicator completeness on the **fact** stream	An **archived** indicator, which serves valid metadata while its data endpoint returns an error envelope. Neither a metadata check nor a row-count floor detects this
Parse	Typed contracts, whitespace and empty-string normalisation, in-batch duplicate detection	A trailing space splitting a group-by; `""` defeating null checks; a source returning a key twice
Pre-load	Domain invariants: key in the configured set, coordinate and year ranges, finite values, referential integrity	An aggregate row double-counting a dimension; NaN, which compares false to everything and defeats equality *and* range tests
Batch	Empty-batch and reject-ratio assertions	A partial load reporting success
CDC transport	Connector and task state, slot lag, **source-to-warehouse parity**	Dropped events, invisible in every other signal
ClickHouse landing	Quarantine view over the event log; consumer exception check	An unattributable key; a silently stalled consumer
dbt	58 tests: `unique`, `not_null`, `relationships`, `accepted_values`, `accepted_range`, plus 5 singular tests	Dedup errors, join fan-out, a resurrected tombstone, an incremental model degraded to an append
DAG gates	Parity, CDC lag, source health, marts reconciliation	The above, re-asserted in the critical path so the guarantee survives `dbt run` without tests

On Great Expectations. The brief permits *"a testing framework of your choice (e.g. pytest, dbt tests, Great Expectations)"*, and I chose dbt tests plus pytest. dbt tests run **inside** the warehouse after transformation and are right for uniqueness, referential integrity and business rules, because they live next to the models they constrain. What they structurally cannot reach is a payload that has not landed yet. Something must occupy that earlier slot; in a team deployment I would put GX there specifically because Data Docs give a non-engineer an auditable record of what was rejected and why. Here the slot is filled by explicit assertions in `ingest/checks.py`, labelled in the code as the gate GX would own, because every check this boundary needs is structural rather than distributional. The place I would reach for GX first in *this* pipeline is the ML feature boundary, where the useful expectations are distributional (null-rate profiles, cardinality bounds, label degeneracy) and genuinely are GX's strength rather than dbt's.

Observability design

Signal	Source	Threshold	What a human does
CDC lag	Debezium heartbeat, via the exporter	> 300s for 2m	Check connector and task state, then `system.kafka_consumers` for a stalled consumer
Pipeline health	`ops.ingest_run`, `ops.layer_counts`	parity < 99% for 10m	Compare topic high watermark against landing count; `make verify` prints both
Data freshness	`raw.cdc_event_log`	> 24h **and** heartbeat healthy	Check the ingestion DAG. Alone, informational
Resource usage	ClickHouse `:9363`, Redpanda `:9644`, natively	parts > 300 for 15m	Check insert batch size, then the partition key
Source health	`pg_replication_slots`	> 1 GiB for 5m	`make drop-slot` if inactive. This threatens the **source** disk
Data quality	quarantine and reject tables	> 0 for 5m	Read the reasons; usually a source schema change

The distinction that matters most is between CDC lag and data freshness, and I got it wrong first and fixed it by measurement. With lag computed per business table, an idle dimension reported 1,805 seconds of "lag" after half an hour in which nothing had changed. Nothing was wrong. Alerting on that pages someone nightly for a table with no news. Lag is now measured against a heartbeat row updated on a fixed 10-second timer, which means connector liveness; per-table freshness is separate and informational, and the freshness alert is explicitly suppressed while the heartbeat is unhealthy, because two alerts for one root cause is how a rota learns to ignore both.

Tools. Prometheus and Grafana, with **no exporter sidecars**: ClickHouse and Redpanda expose Prometheus metrics natively, so the usual exporter containers are unnecessary. Two fewer containers, two fewer reasons a dashboard is empty. The one exporter covers only the data-plane signals that live in tables. It queries at **scrape time** rather than caching, because a cached value lets Prometheus stamp a fresh timestamp on stale data; and a failed query yields **no metric** rather than a zero, because zero is a plausible row count while an absent series makes a gap and fires `absent()`. Grafana is provisioned from the repository with **no plugins**, because the ClickHouse datasource plugin would make the container's start depend on reaching grafana.com.

Alert rules are code, and untested rules fail silently, so all 10 have `promtool` tests covering firing *and* staying quiet. `CdcParityBroken` is why: it divided two series carrying different label names, and PromQL matches on the full label set, so the expression evaluated to an empty vector and the alert

could never have fired, while looking exactly like one that was passing.

The same lesson then recursed one level up. Auditing this report's claim that every rule was tested found that **four of the ten had no test at all**, and `promtool` had been reporting SUCCESS throughout, because it only runs the cases it is given and cannot know which rules were never named. The suite was silently incomplete in precisely the way it exists to prevent. The gap is closed: 13 cases now cover all 10 rules, verified by comparing the alertnames the tests reference against the rules defined rather than by trusting the exit code.

Alertmanager routing is deliberately not wired: it adds a container and credentials to prove something invisible in a ten-minute review. Every rule carries a runbook instead.

CDC operational failure modes

Added section. These are what separate a CDC pipeline that survives production from one that works on a laptop, and the design above is shaped by them.

1. The forgotten replication slot. An inactive logical replication slot retains WAL indefinitely so a consumer that might return can resume. If none returns, WAL accumulates until the **source** database's disk fills. What makes it hard to catch is that there is nothing unhealthy to look at, in either of the two ways it happens: the connector is gone entirely, so there is no consumer to inspect, or a connector is still running perfectly well against a newer slot while an orphaned one retains WAL behind it. Three controls: `max_slot_wal_keep_size=2GB` so the server invalidates the slot rather than filling its disk; a heartbeat so the slot advances even when the captured tables are idle; and `make down`, which removes the connector before dropping the slot, in that order.

I have recovered a managed PostgreSQL cluster where three abandoned replication slots were each retaining hundreds of GB of WAL, and it was the second of those two cases. Nothing surfaced it: I found them while auditing active sessions on the cluster, and the connectors still running looked fine. The expensive part was not the diagnosis. I dropped the slots and reclaimed the space, and the running connectors simply recreated them, so the WAL began accumulating again.

That is the whole reason `make down` removes the connector first and refuses to drop a slot that is still attached rather than failing quietly. Dropping the slot treats the symptom; the consumer is what keeps making a new one.

2. Dedup ordering, when a log position is compared as text. Any "latest version wins" rule needs a total order over log positions, and the trap is that the obvious ordering is a string comparison. That applies directly to this pipeline. A Postgres LSN is rendered as hex with a slash, and compared as text `0/10000000` sorts **before** `0/9FFFFFF` even though it is the later position, so a dedup keyed on the text form would elect a stale row for any key with events either side of a digit-count boundary. This design avoids it by construction rather than by care: Debezium emits `__lsn` as a JSON integer, the materialized views extract it with `JSONExtractUInt` into a `UInt64`, and `ReplacingMergeTree` therefore compares versions arithmetically.

I have hit this class of bug in production, in a MySQL-based pipeline rather than a Postgres one, where the dedup ordering key was the binlog file name compared as a string: at the rollover from `mysql-bin.999999` to `mysql-bin.1000000` the comparison inverts, and for any key with events on both sides of that boundary the older row won, silently. The fix was to sort on the numeric suffix of the file name rather than on the name. It is why the version column here is a number before it is anything else, and why I now treat "what is the total order over log positions" as a question to answer explicitly rather than a default to inherit.

3. Logical decoding carries no DDL. Postgres streams row changes, not schema changes, so schema drift surfaces **in the data**. This design absorbs that: `JSONAsString` plus `JSONExtract` means an unexpected payload produces nulls rather than a stalled consumer, and the immutable log retains the original payload so a model can be re-derived once the schema is understood. Without a Schema Registry there is no wire-level evolution contract, and that is the first thing I would add.

4. Snapshot memory pressure. An initial snapshot streams a whole table through the connector, and a buffered read will exhaust a modestly sized worker. A non-issue at this volume; the first thing to size for at production volume.

Scaling and extending for increasing data volume

Three horizons, each with a **numeric trigger** rather than a vague "at scale".

Horizon 1, the same shape at 100M rows and beyond. Worth being honest that this source cannot reach it: the entire World Bank catalogue at monthly grain, roughly 200 countries by 500 indicators by 66 years by 12, is about 80M rows, and the configured slice is 2,970. The volume that triggers this horizon comes from a different source shape, and for a MERL deployment the obvious one is ERP change data plus survey submissions, where the row count is transactional rather than statistical. What changes past ~100M rows per table: adopt monthly `PARTITION BY toYYYYMM(...)` so expiry and backfill drop whole parts; switch the staging views to incremental tables once `FINAL` at query time becomes measurable; raise topic partitions and `kafka_num_consumers` together, since ClickHouse consumer parallelism is bounded by partitions; and move the fact model from `delete+insert` to `insert_overwrite` on the partition, avoiding mutations entirely.

Horizon 2, more sources and durability. A Schema Registry with Avro, so schema evolution has a wire-level contract instead of being discovered in the data. `ReplicatedMergeTree` with ClickHouse Keeper and a `Distributed` table; note the dbt adapter's distributed materializations are experimental and read-after-write on a cluster needs care. Per-source connectors, so a slow source cannot hold up another's slot. Airflow moves to KubernetesExecutor with one pod per task, and the DAG becomes a factory over a source manifest, so a new source is a config entry. Alertmanager routing on the existing rules.

One trap in this direction is worth naming, because it makes the platform look more production-ready while making CDC less reliable: **a logical replication slot does not survive a Postgres failover.** Put the source on a three-instance HA cluster and a switchover leaves the connector's slot on the old primary, so it silently loses its position and either gaps or re-snapshots. PostgreSQL 17 added slot synchronisation to standbys, and it has to be configured and then tested by actually killing the primary, because a backup job reporting success is not evidence that a slot survived. The single-node source here sidesteps the problem entirely, which is the honest reason it is single-node rather than an oversight.

Horizon 3, what changes qualitatively. Below ~10M rows per table this design is over-engineered and a scheduled batch copy would do. Above ~1B the single-node assumption breaks and the interesting problems become shard-key selection and whether the mart should be pre-aggregated rather than recomputed. The read-acceleration ladder, in climbing order: sort-key refinement, projections, dictionaries for dimension lookups, `AggregatingMergeTree` rollups, then tiered storage with TTL moves. Building any of them at 2,970 rows would be theatre.

The cheapest axis, already built. Adding indicators is a config change in `ingest/indicators.yml`, because the fact is keyed on `(country, indicator, year)`. A new series is more rows, never more columns.

Trade-offs, and what I would do with more time

Chosen	Rejected	Why, and what it costs
Redpanda	Kafka + ZooKeeper	One container instead of three, `rpk` for demos, Kafka-API compatible. Cost: not the exact runtime named in the brief
ClickHouse Kafka engine + views	`clickhouse-kafka-connect` sink	No custom Connect image or plugin install, and it produces real materialized views to reason about. Cost: weaker per-connector observability than Connect's REST API and DLQ
Plain JSON on the wire	Schema Registry + Avro	One fewer container, topics readable with `rpk`. **Cost: no wire-level schema-evolution contract.** First thing to add
dbt tests + pytest	Great Expectations	Structural checks, cheaper and more precise here, and unit tested. Cost: no Data Docs artifact for non-engineers
Current-state marts	SCD2 history	Matches the source, which is itself current-state. The 30-day event log means history is reconstructable with a dbt snapshot
`airflow standalone`, 2 containers	Split webserver/scheduler/worker	Fits alongside a broker, a JVM and two databases on one laptop. Cost: a demo topology, not production. The production one is described above rather than half-built

With more time, in priority order: a Schema Registry, the one missing piece with a real correctness consequence; a GX suite at the ML feature boundary; the `ReplicatedMergeTree` migration, since the path is written down; dbt snapshots over the event log to reconstruct history; per-source connector isolation.

What I would most want to talk through is where the boundary should sit between the warehouse and the feature layer. `agg_country_year_features` carries completeness and series vintage on each row so a model author can see what they are training on, and I am genuinely unsure whether that belongs there or in a feature store with point-in-time joins.

The thing I found most interesting here was smaller and more specific. Measuring CDC lag per table looked obviously correct and was wrong: an idle dimension reports the same number as a stopped connector, so the signal cannot distinguish "nothing changed" from "nothing is arriving". Splitting connector liveness from data freshness took about ten minutes and changed which of the two signals I would act on.

Deliberate scope boundaries are recorded in `docs/SCOPE.md`, frozen before any code was written: Great Expectations; Schema Registry and Avro; ClickHouse clustering and sharding; projections, dictionaries, rollups and TTL beyond the ops tables; SCD2 history; Alertmanager routing; any BI layer, Terraform or Kubernetes manifests. A stated omission with a reason is a decision; a silent one is a gap.